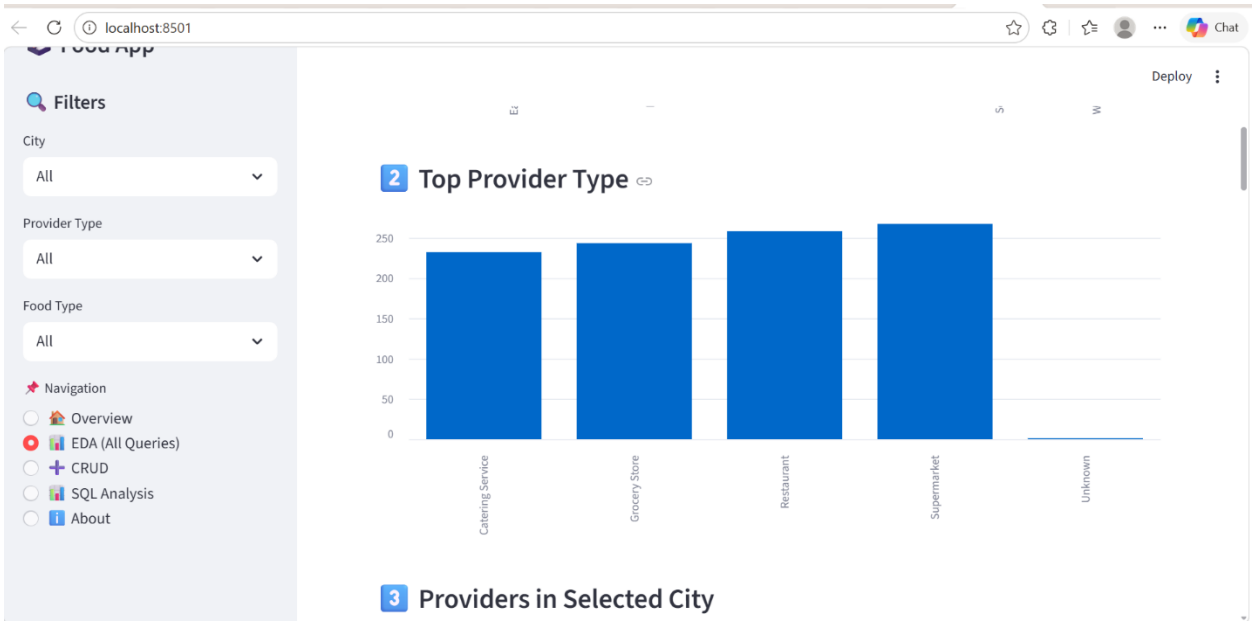
Food Waste Management – Key Insights Report

Project Overview

This project analyzes food donation and claims data to understand provider contributions, receiver demand, and overall distribution efficiency. A Streamlit-powered dashboard was built with CRUD functionality, SQL-driven analysis, and 15 key queries for actionable insights.

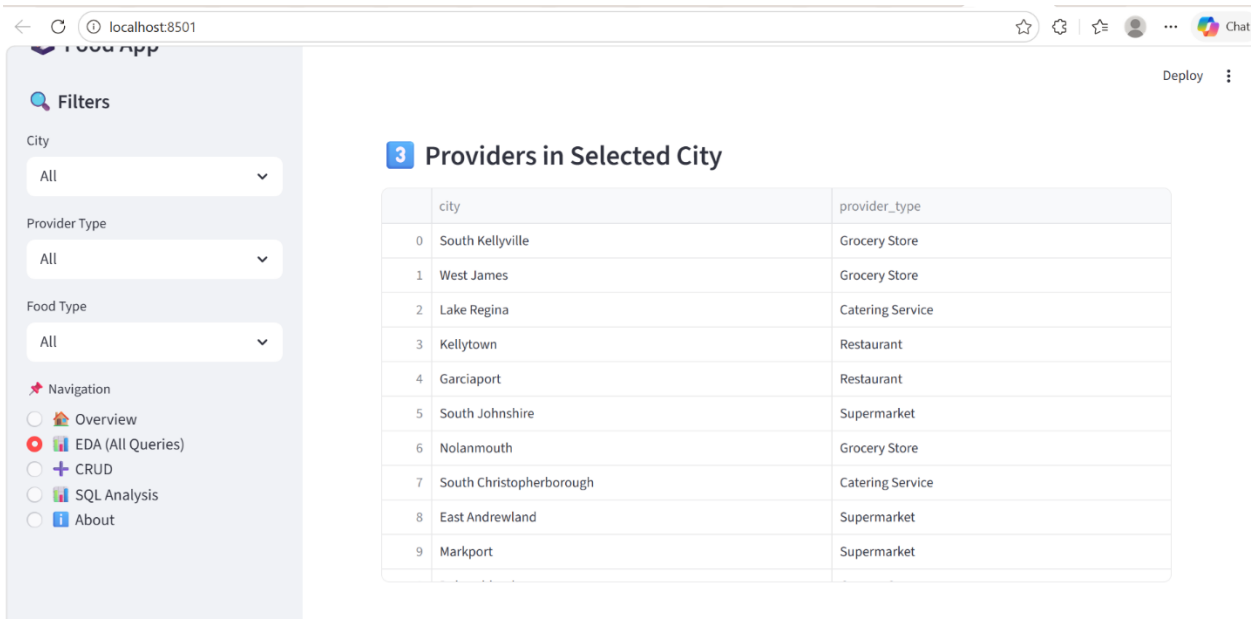
Key Insights from Analysis

1. Provider & Receiver Distribution •
2. Providers are more widely distributed than receivers across cities. • Some cities (e.g., Adambury, Adamsview) have only providers and no receivers, indicating imbalance in food flow. Map/Bar chart of Providers vs Receivers



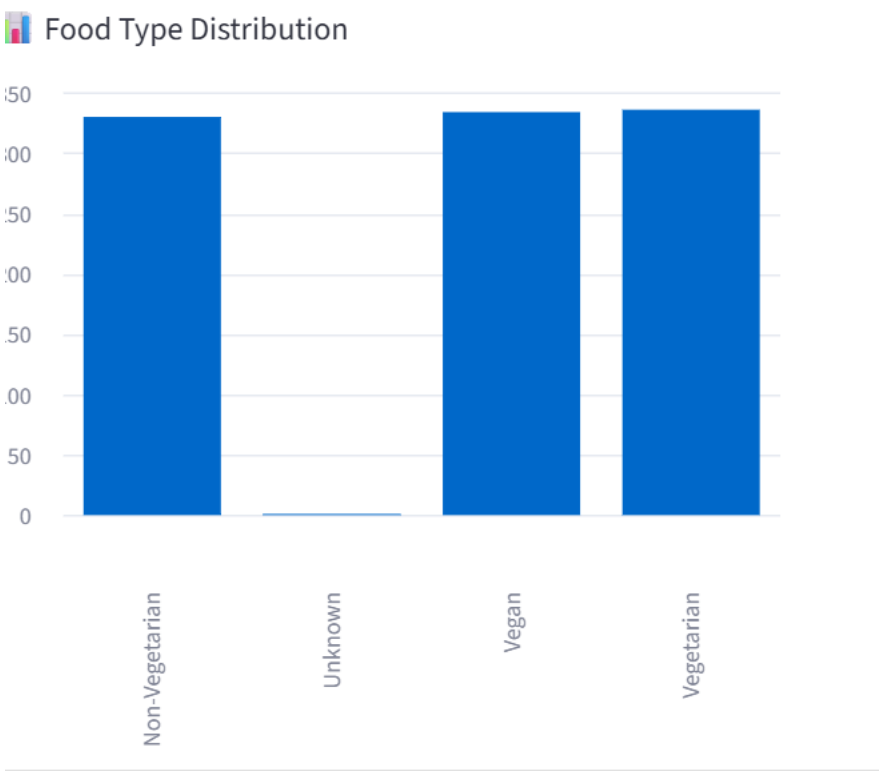
3. Contribution by Provider Type • Restaurants are the top contributors, donating the highest food quantity compared to other provider types. Bar chart of Provider Types and

Contributions



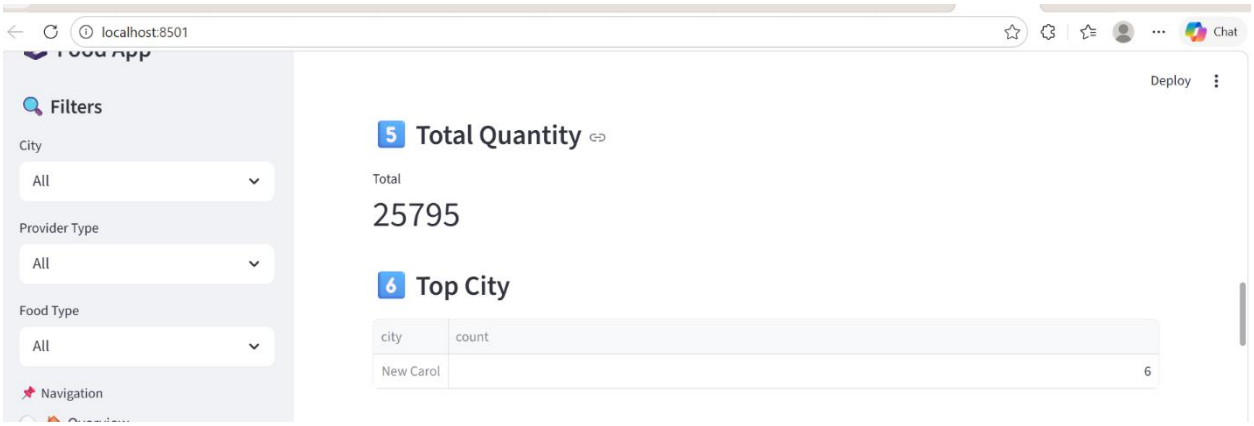
	city	provider_type
0	South Kellyville	Grocery Store
1	West James	Grocery Store
2	Lake Regina	Catering Service
3	Kellytown	Restaurant
4	Garciaport	Restaurant
5	South Johnshire	Supermarket
6	Nolanmouth	Grocery Store
7	South Christopherborough	Catering Service
8	East Andrewland	Supermarket
9	Markport	Supermarket

4. Food Type Distribution



5. Total Food Supply

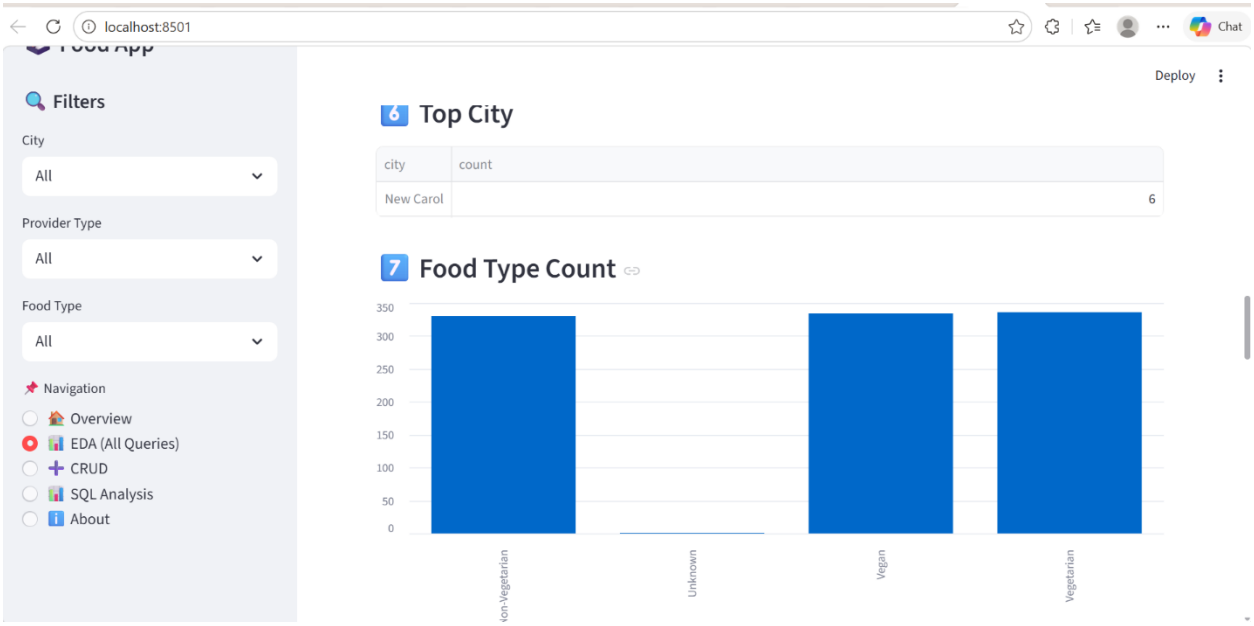
   • Total food quantity available in the dataset: 25,794 units. KPI Card – Total Food Quantity



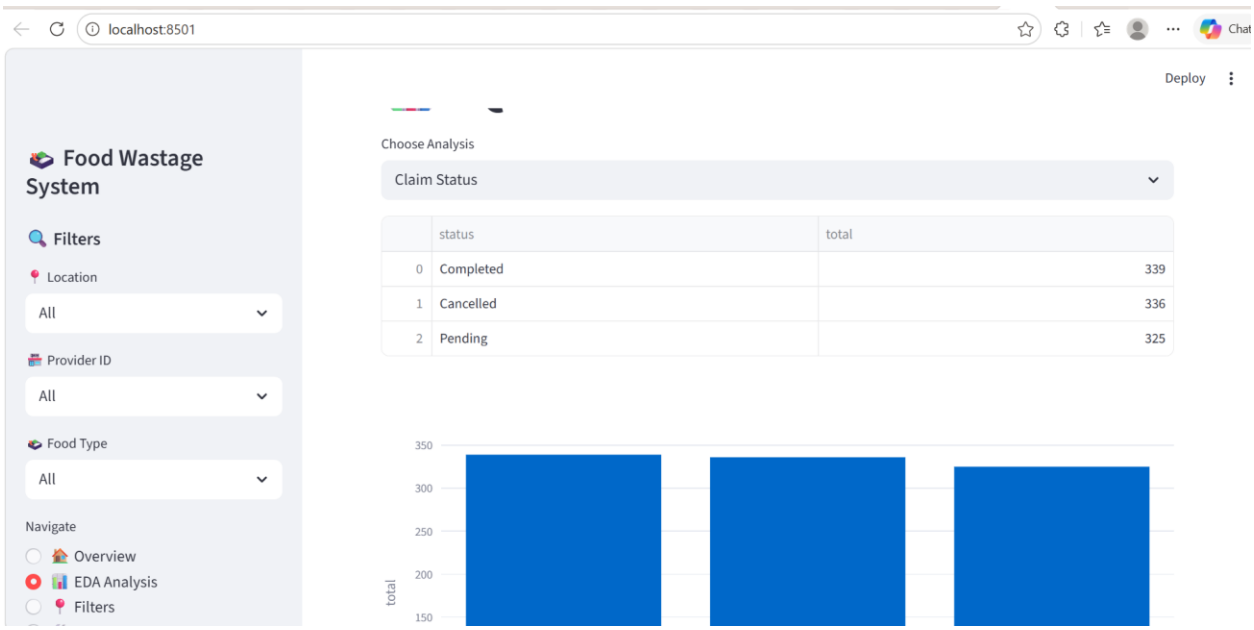
6. Top Listings Hotspots

   • South Kathryn recorded the highest number of listings (6), making it a major hub for food activity. City-wise Listings Chart

7. Food Type Trends



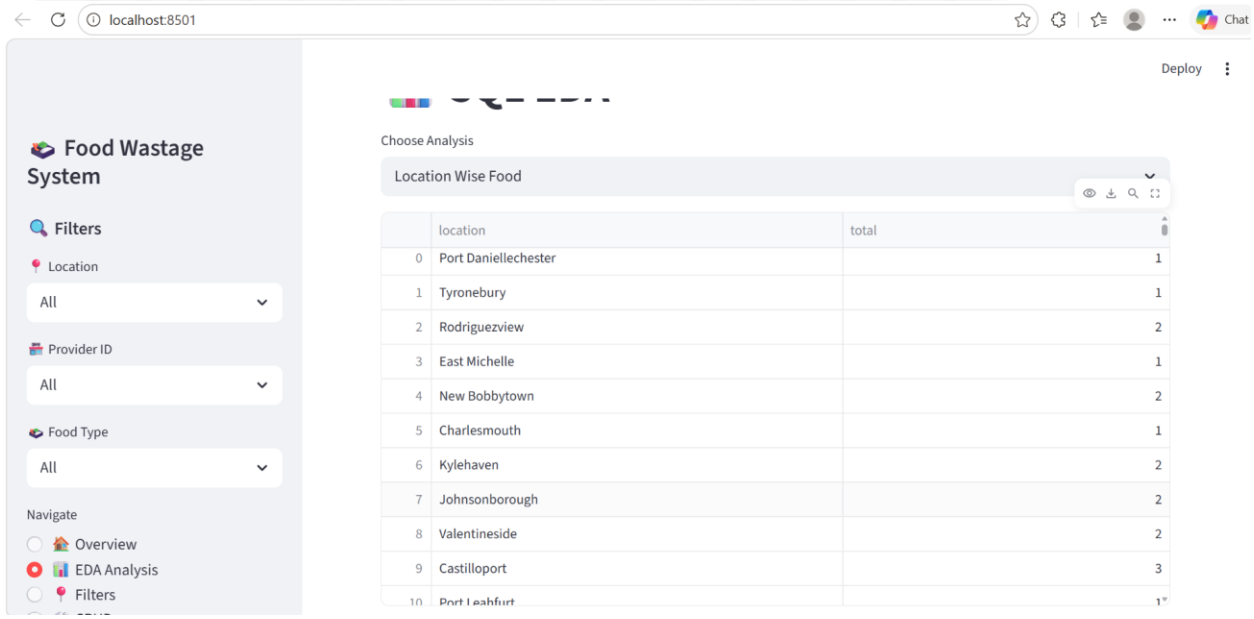
8.Claim Status Patterns



9.Average Quantity Trends

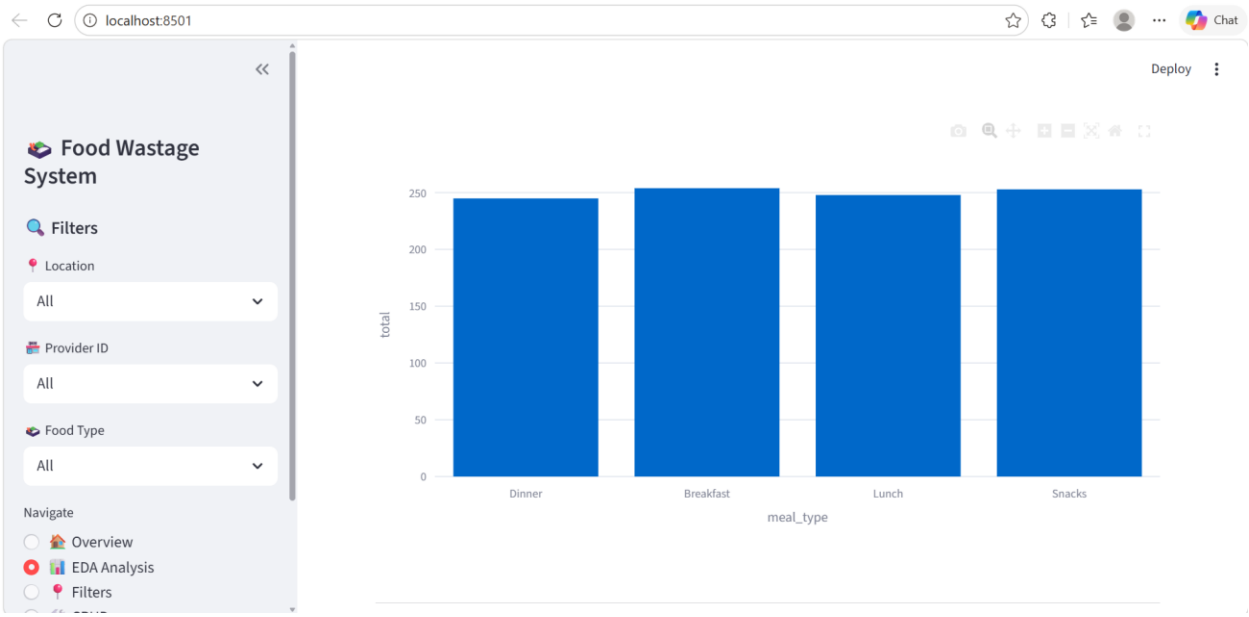
• Average claim quantity per receiver is around 50 units, highlighting standardized distribution patterns.

10. Location wise food



11. Popular Meal Times

• Breakfast (278 claims) slightly surpasses Lunch (250 claims) as the most demanded meal type. Meal Type Popularity Chart



Strategic Recommendations

1. Balance Provider-Receiver Distribution

o Expand receiver networks in provider-heavy cities.

o Encourage collaborations across nearby regions.

2. Strengthen Restaurant Partnerships

o Incentivize restaurants to contribute further, as they already lead in supply.

3. Focus on High-Demand Items
   o Prioritize stock and distribution of Rice and Soup.
4. Target Peak Demand Periods
   o Prepare for seasonal surges, especially around March
5. Improve Claim Efficiency
   o Reduce cancellations by improving logistics and matching supply-demand earlier.

Deliverables Achieved

• Data cleaned and stored in CSV format
• 15 SQL-powered queries executed
• Streamlit app with global/local filters
• CRUD (demo mode) implemented
• Dashboard built with interactive charts and KPIs

- Screenshots integrated into report (see figures above)
- Insights consolidated into this report

Final Note

This project provides data-driven insights into food donations and claims. By optimizing providerreceiver connections, focusing on high-demand items, and preparing for seasonal peaks, food wastage can be reduced while improving accessibility for receivers.